

Normal Forms

Database course - Dr/ Yasser
by Mahmoud Azmy - Cyber/G2



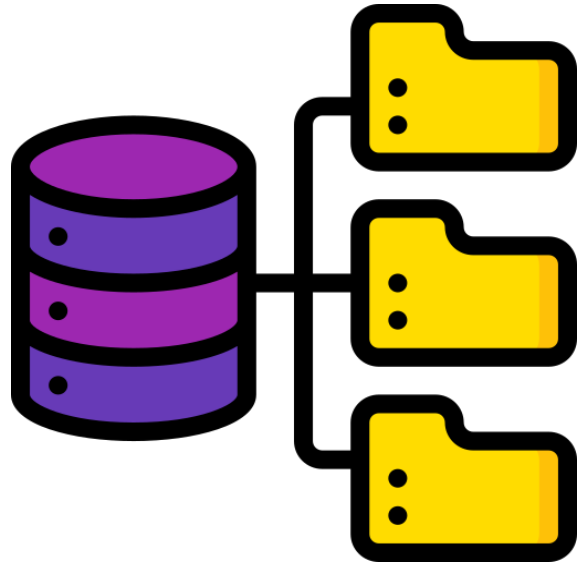
What is Normalization?

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

Eliminates duplicate data

Ensures data dependencies make sense

Prevents update, insertion, and deletion anomalies



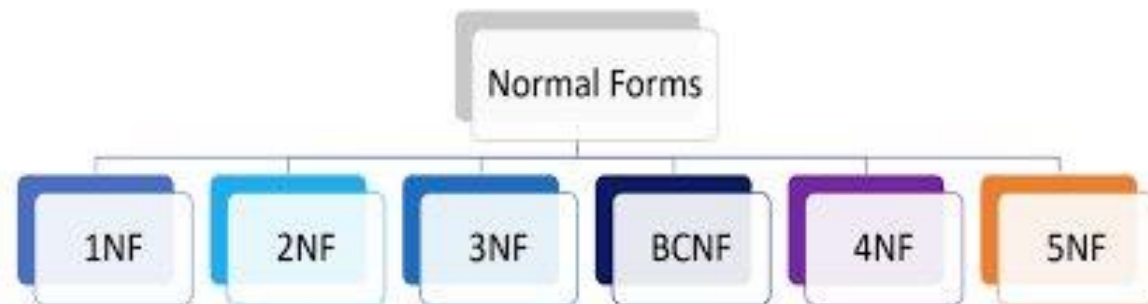
Why Normalization Matters

Benefits:

- ✓ Reduces data redundancy
- ✓ Improves data consistency
- ✓ Makes database maintenance easier
- ✓ Prevents logical inconsistencies



Without proper normalization, updating data can lead to errors and inconsistencies



Recap: First Normal Form (1NF)

- **Must have a Primary Key**
 - Row order cannot convey information.
- Each column must contain only one data type.
 - Table must have a primary key.
- No repeating groups allowed.

Recap: Second Normal Form (2NF)

Remove Partial Dependencies

Each non-key attribute must depend on the **entire primary key**, not just part of it.

This rule applies when you have a composite primary key (multiple columns).

Recap: Third Normal Form (3NF)

Remove Transitive Dependencies

Each non-key attribute must depend on the key, the whole key, and nothing but the key.

Non-key attributes should not depend on other non-key attributes.

Recap: Boyce-Codd Normal Form (BCNF)

All attributes depend on the key, the whole key, and nothing but the key

A stronger version of 3NF where **every** attribute (including key attributes) must depend properly on candidate keys.

Eliminates all anomalies related to functional dependencies.

Introduction to Higher Normal Forms

Beyond BCNF, we have 4NF and 5NF that deal with more complex dependency issues.

4th Normal Form (4NF)

Addresses multivalued dependencies

5th Normal Form (5NF)

Addresses join dependencies

Fourth Normal Form (4NF) - Definition

Rule- define :

The only kinds of multivalued dependencies allowed in a table are multivalued dependencies on the key.

In other words: Remove multivalued dependencies that don't depend on the entire key.

Problem: House design website with models, colors, and styles all in one table.

Model
"Villa"
available Colores
"yellow" & "blue"
available styles
"modern" & "classic"

Model
"Studio"
available Colores
"brown" & "grey"
available styles
"Minimalist" & "neoclassical"

Model
"Apartment"
available Colores
"brown" & "beige"
available styles
"modern" & "Rustic"

Before 4NF (All columns are key):

<u>Model</u>	<u>Colores</u>	<u>Style</u>
Villa	yellow	modern
Villa	yellow	classic
Villa	blue	modern
Villa	blue	classic
Studio	brown	Minimalist
Studio	brown	neoclassical
Studio	grey	Minimalist
Studio	grey	neoclassical
Apartment	brown	modern
Apartment	brown	Rustic
Apartment	beige	modern
Apartment	beige	Rustic



Issue: Adding a new color (e.g. **RED**) for Apartment requires multiple rows.
Colors and styles are independent but forced together.

4NF - Solution

Solution: Split into two tables where each multivalued dependency is on the key.

<u>Model</u>	<u>Styel</u>
Villa	modern
Villa	classic
Studio	Minimalist
Studio	neoclassical
Apartment	modern
Apartment	Rustic

<u>Model</u>	<u>color</u>
Villa	yellow
Villa	blue
Studio	brown
Studio	grey
Apartment	brown
Apartment	beige
Apartment	RED

Each table represents one independent fact. Join them to find what products people will eat. No redundancy, no inconsistency!

Fifth Normal Form (5NF) - Definition

Rule - definition:

It must not be possible to describe the table as being the logical result of joining some other tables together.

In other words: If a table can be reconstructed by joining smaller tables, it should be split into those smaller tables.

Problem: Ice cream preferences - brands offer flavors, people like brands, people like flavors.

Brand "Azza"
available Flavors
"Vanilla"
"coco"
"strawberry"
"mango"

Brand "B'laban"
available Flavors
"Vanilla"
"Pistachio"

Brand "Sultana"
available Flavors
"Vanilla"
"strawberry"
"mango"

"i love "vanilla" & "coco", from
"Azza" & "B'laban" "

"i love "Pistachio", "mango" &
^Strawberry^, from ^B'laban^
& ^Sultana^"

Before 5NF (Deduced combinations):

Person	Brand	Flavor
Man	Azza	Vanilla
Man	Azza	coco
Man	B'laban	Vanilla
Woman	B'laban	Pistachio
Woman	Sultana	mango
Woman	Sultana	Strawberry



Issue: If Woman now likes Azza brand, we must add multiple rows (one for each flavor she likes that Azza offers). Easy to create inconsistencies!

5NF - Solution

Solution: Split into three independent tables representing the three facts.

<u>Person</u>	<u>Brand</u>
Man	vanilla
Man	coco
Woman	Pistachio
Woman	mango
Woman	strawberry

<u>Person</u>	<u>Brand</u>
Man	Azza
Man	B'laban
Woman	B'laban
Woman	Sultana

<u>Brand</u>	<u>flavor</u>
Azza	Vanilla
Azza	coco
Azza	strawberry
Azza	mango
B'laban	Vanilla
B'laban	Pistachio
Sultana	Vanilla
Sultana	strawberry
Sultana	mango

Each table represents one independent fact. Join them to find what products people will eat. No redundancy, no inconsistency!

4NF vs 5NF - Quick Comparison

4th Normal Form

Deals with multivalued dependencies

Two independent sets of values

Solution: Split into 2 tables

Example: Model has colors

AND styles independently

5th Normal Form

Deals with join dependencies

Three or more independent facts

Solution: Split into 3+ tables

Example: Brands offer flavors,
people like brands, people like
flavors

Any Questions?
,Thanks for
your time!

